

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital Signal and Image Processing(01CT1513)	Aim :- Perform gray level operations images.	
Lab-7	14-08-2026	Enrolment No:92400133131

Ami :- Perform gray level operations images.

```

Code :- import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image in grayscale
src_image = cv2.imread(
    r"1.JPJ.jpg",
    cv2.IMREAD_GRAYSCALE
)

# Check if image is loaded
if src_image is None:
    print("Image not found!")
    exit()

# 1. Image Negation
negative_image = 255 - src_image

# 2. Thresholding
_, thresholded_image = cv2.threshold(
    src_image,
    128,
    255,
    cv2.THRESH_BINARY
)

# 3. Gamma Correction
gamma = 2.0

# Normalize image to range [0, 1]
normalized_image = src_image / 255.0

# Apply gamma correction
gamma_corrected_image = np.power(
    normalized_image,
    1 / gamma

```

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)

# Convert back to range [0, 255]
gamma_corrected_image = np.uint8(
    gamma_corrected_image * 255
)

# Display all images
plt.figure(figsize=(15, 5))

plt.subplot(1, 4, 1)
plt.imshow(src_image, cmap="gray")
plt.title("Original")
plt.axis("off")

plt.subplot(1, 4, 2)
plt.imshow(negative_image, cmap="gray")
plt.title("Negative")
plt.axis("off")

plt.subplot(1, 4, 3)
plt.imshow(thresholded_image, cmap="gray")
plt.title("Threshold")
plt.axis("off")

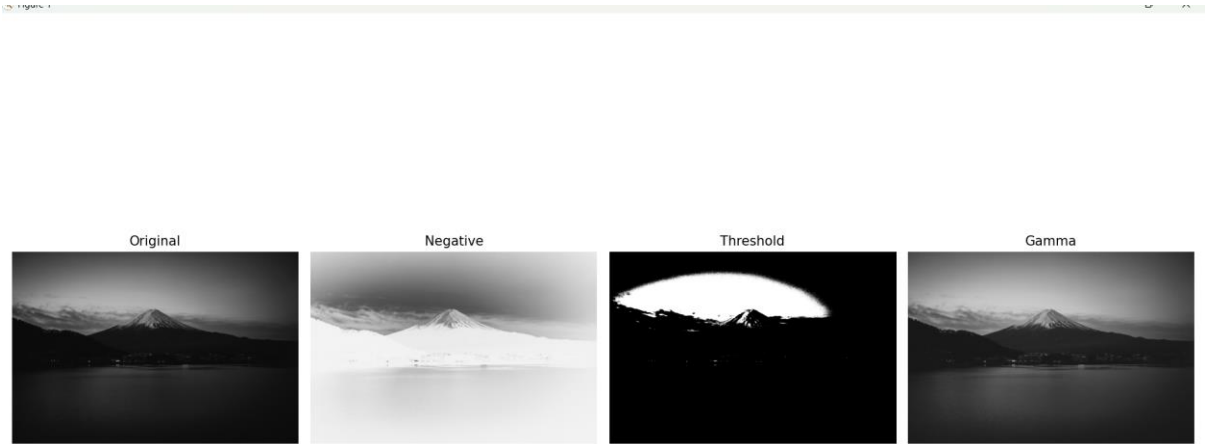
plt.subplot(1, 4, 4)
plt.imshow(gamma_corrected_image, cmap="gray")
plt.title("Gamma")
plt.axis("off")

plt.tight_layout()
plt.show()

```

OUTPUT :-

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GAMMA =0.5

